

Auspicious Tatami Mat Arrangements

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Abstract. We introduce tatami tilings, and present some of the many interesting questions that arise when studying them. Roughly speaking, we are considering tilings of rectilinear regions with 1×2 dimer tiles and 1×1 monomer tiles, with the constraint that no four corners of the tiles meet. Typical problems are to minimize the number of monomers in a tiling, or to count the number of tilings in a particular shape. We determine the underlying structure of tatami tilings of rectangles and use this to prove that the number of tatami tilings of an $n \times n$ square with n monomers is $n2^{n-1}$. We also prove that, for fixed-height, the number of tatami tilings of a rectangle is a rational function and outline an algorithm that produces the coefficients of the two polynomials of the numerator and the denominator.

Keywords: dimer, monomer, tatami, tilings, combinatorics, enumeration.

1 What Is a Tatami Tiling?

Traditionally, a tatami mat is made from a rice straw core, with a covering of woven soft rush straw. Originally intended for nobility in Japan, they are now available in mass-market stores. The typical tatami mat occurs in a 1×2 aspect ratio and various configurations of them are used to cover floors in houses and temples. By parity considerations it may be necessary to introduce mats with a 1×1 aspect ratio in order to cover the floor of a room. Such a covering is said to be “auspicious” if no four corners of mats meet at a point. Hereafter, we only consider such auspicious arrangements, since without this constraint the problem is the classical and well-studied dimer tiling problem [1], [2]. Following Knuth [3], we will call the auspicious tatami arrangements, *tatami tilings*. The enumeration of tatami tilings that use only dimers (no monomers) was solved in [4].

Perhaps the most commonly occurring instance of tatami tilings is in paving stone layouts of driveways and sidewalks, where the most frequently used paver has a rectangular shape with a 1×2 aspect ratio. Two of the most common patterns, the “herringbone” and the “running bond”, have the tatami property (see Fig. 1). Consider a driveway of the shape in Fig. 1. How can it be tatami tiled with the least possible number of monomers? The answer to this question could be interesting both because of aesthetic appeal, and because it could save work, since to make a monomer a worker typically cuts a 1×2 paver in half.

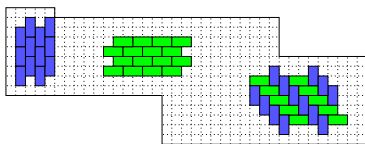


Fig. 1. The dimers shown illustrate, from left-to-right, the vertical running bond pattern, the horizontal running bond pattern, and the herringbone pattern. Ignoring those patterns, what is the least number of monomers among all tatami tilings of this region? The answer is provided at the end of the paper (Fig. 6).

Before attempting to study tatami tilings in general orthogonal regions it is crucial to understand them in rectangles, and our results are about tatami tilings of rectangles. Here is an outline of the paper. In Section 2 we determine the structure of tatami tilings in a rectangle. This structural characterization has important algorithmic implications, for example, it reduces the size of the description of a tiling from $O(rc)$ to $O(\max\{r, c\})$ and may be used to generate tilings quickly. In Section 3 we provide some counting results for tatami tilings in a rectangle: The number of tilings of an $n \times n$ square with n monomers is $n2^{n-1}$ and for a fixed number of rows r , the ordinary generating function of the number of tilings of an $r \times n$ rectangle is a rational function. In Section 4 we return to the question of tatami tiling general orthogonal regions and introduce the “magnetic water strider problem”. Additional conjectures and open problems are also introduced in this section.

2 The Structure of Tatami Tilings: T-Diagrams

We show that all tatami tilings have an underlying structure which partitions the grid into blocks, where each block is filled with either the vertical or horizontal running bond. We describe this structure precisely and prove some results for rectangular regions.

Notice that in Fig. 2(a), wherever a horizontal and vertical dimer share an edge, the placement of another dimer is forced to preserve the tatami condition. These herringbone formations, called *rays*, propagate themselves to the boundary of the grid and do not intersect one another. Between the rays, there are only vertical or horizontal running bonds. Wherever a running bond meets the grid boundary, either a *smooth* or *jagged* edge is formed. Jagged describes edges in which dimers alternate with monomers and otherwise an edge is smooth.

In a rectangular grid, a ray may be forced by one of four possible *sources*. To discover what they are we consider the tiles that can cover the grid squares at the beginning of a ray. Referring to the inline diagrams, the innermost square  may not be covered by a vertical dimer  because this would extend the ray. If it is covered by a horizontal dimer , this is called a *hamburger*. Otherwise it is covered by a monomer  in which case we consider the grid square beside

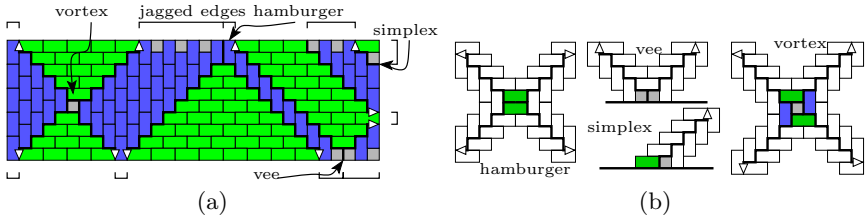


Fig. 2. (a) A tatami tiling whose **T**-diagram contains all four types of features, showing jagged edges. (b) The four types of features with their local tilings.

it $\begin{smallmatrix} \square & \square \\ \blacksquare & \square \end{smallmatrix}$. If it is covered by a monomer the source is called a *vee* $\begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix}$, by a vertical dimer the source is called a *vortex* $\begin{smallmatrix} \square & \square \\ \blacksquare & \blacksquare \end{smallmatrix}$, and if it is covered by a horizontal dimer it is called a *simplex* $\begin{smallmatrix} \blacksquare & \square \\ \square & \square \end{smallmatrix}$. Each of these four sources forces at least one ray in the tiling and all rays begin at either a hamburger, vee, vortex or simplex. These source-ray combinations are depicted in Fig. 2(b).

The set of bold curves in each of the four source-ray drawings in Fig. 2(b), is called a *feature*. Just as rays cannot intersect, neither can features in a tatami tiling, so a *feature-diagram* refers to a set of non-intersecting features drawn in a grid. A feature-diagram admits a tatami tiling if the grid can be tiled so that the tiles propagating from different features agree, as depicted in Fig. 2(b). In this case it is called a **T**-diagram. A feature diagram is a **T**-diagram if and only if rays bounding the same block admit compatible bond patterns and the distance between them has the correct parity. Some incompatible configurations and distance parities are shown in Figures 3(c) and 3(d).

This characterization guarantees a verification that a given feature-diagram is a **T**-diagram, which is linear in the perimeter of the grid. Furthermore, a Tatami tiling can be validated using only the information at its boundaries (see Figures 3(b) and 3(a)).

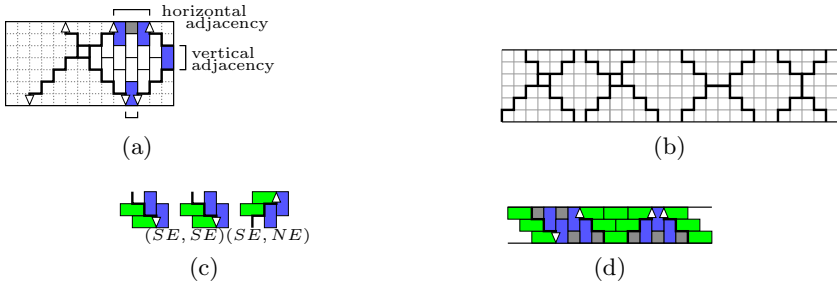


Fig. 3. (a) Fill a contiguous block of a **T**-diagram with the bond pattern by laying along the boundaries first and then filling from the outside inward. (b) Hamburgers and vortices can be differentiated by their endpoints. (c) Incompatible “adjacent” rays. (d) Distance parities between rays generalize inductively from these.

3 Counting Results

Let $T(r, c, m)$ be the number of tatami tilings of a rectangular grid with r rows, c columns, and m monomers. Also, $T(r, c)$ will denote the sum

$$T(r, c) = \sum_{m \geq 0} T(r, c, m).$$

We begin by giving necessary conditions for $T(r, c, m)$ to be non-zero.

Theorem 1. *If $T(r, c, m) > 0$, then m has the same parity as rc and $m \leq \max(r + 1, c + 1)$.*

Proof. Let r, c and m be such that $T(r, c, m) > 0$ and let d be the number of grid squares covered by dimers in an $r \times c$ tatami tiling so that $m = rc - d$. Since d is even, m must have the same parity as rc .

We may assume that $r \leq c$, and prove that $m \leq c + 1$. The first step is to prove that a monomer on a side boundary of any tiling can be mapped to the top or bottom. We then observe that for any pair of consecutive monomers on the top or bottom boundary, there is a horizontal dimer on the opposite boundary, and for any vortex there are horizontal dimers on each of the top and bottom boundaries. With this it is possible to show the bound on the number of monomers.

Let T be a tatami tiling of the $r \times c$ grid with a monomer on the West boundary, touching neither the South nor the North boundary. Such a monomer is on a jagged edge of a block of horizontal dimers. Define a *diagonal slice* to be a set of dimers in this block which form a stairway from this monomer to either the top or bottom of the grid, as in Fig. 4. If such a diagonal slice exists it can be flipped diagonally, which changes the orientation of its dimers and maps the monomer to the other end of the slice. Suppose neither slice exists, then they must each be impeded by a distinct ray. Such rays have this horizontal block to the left so one of them is directed SE and the other NE and they meet the right side before intersecting. Referring to Fig. 4, $\alpha + \beta + 3 = \gamma + \delta + 1 \leq r$ while $c \leq \alpha + \gamma = \beta + \delta$ and $r \leq c$. Thus $\alpha + \beta + 3 \leq \alpha + \gamma$ implying that $\beta < \gamma$. But at the same time, $\gamma + \delta + 1 \leq \beta + \delta$ implies that $\gamma < \beta$, which is a contradiction. Therefore one of the diagonal slices exists and the monomer can be moved.

Assume that no monomers are strictly on the sides of the tiling, so that all monomers are either on the top or bottom boundaries or in vortices. Let α be the number of vortices.

Consider a vee on the North boundary and the block of horizontal dimers directly South. This block must reach the South boundary, otherwise, by an argument similar to the above, we would have $c < r$. Both above and below any vortex, there are blocks of horizontal dimers which reach the top and bottom boundaries. More precisely, there is an injective function from the set of vees and vortices² to horizontal monomers on the North or South boundaries.

Define two binary sequences, one for the top row of grid squares and the other for the bottom row, where there is a 1 for each square covered by a monomer

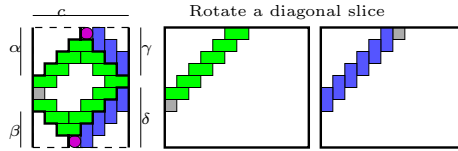


Fig. 4. If both diagonal slices are impeded then there cannot be as many columns as rows. Note that vees are implicitly impossible.

and 0s elsewhere. There are α distinct pairs of 0s in each sequence, one for each vortex, so we remove a 0 from each pair. Now we have two sequences, each of length $c - \alpha$ and there is an injective function from consecutive pairs of 1s to consecutive pairs of 0s. For each pair of 1s, remove one of the corresponding 0s and squeeze it between the pair of 1s. There are no two adjacent 1s in the new sequences, whose lengths shall be called s and t , and we have $s + t = 2c - 2\alpha$. The total number of 1s is at most $\lceil \frac{s}{2} \rceil + \lceil \frac{t}{2} \rceil$, which is at most $c - \alpha + 1$ implying that there are at most $c + 1$ monomers in total, adding back in the α monomers in vortices. This is all that was required because we proved earlier that the number also has the same parity as rc .

The maximum is achieved in a bond pattern. □

The converse of Theorem 1 is false. For example $T(9, 13, 1) = 0$.

3.1 Square Tatami Tilings

In this Section, we count $T(n, n, n)$, the number of tilings of an $n \times n$ square with n monomers. The number of these for each n is a surprisingly nice sequence. We will frequently refer to flipping diagonal slices of a tiling, which is as described in Theorem 1, or, equivalently, to flipping a monomer in a particular direction. We give the following Lemmas without proof since they are intuitively true.

Lemma 1. *A tiling of a rectangle has the maximum number of monomers for its dimensions if and only if it has no hamburgers and no vortices. For an $n \times n$ square, the maximum number of monomers is n .*

Lemma 2. *Every tiling of an $n \times n$ square with n monomers can be generated from a single trivial tiling (up to rotational symmetry) by flipping diagonal slices. Further, every such tiling has exactly two monomers in corners and they are in adjacent corners.*

Theorem 2. *The number of $n \times n$ tilings with n monomers, $T(n, n, n)$, is $n2^{n-1}$.*

Proof. We will count the $n \times n$ tilings with n monomers up to rotational symmetry by fixing two monomers in the northern corners. Thus, let $S(n) = T(n, n, n)/4$. We will give a combinatorial proof that $S(n)$ satisfies the following recurrence:

$$S(n) = 2^{n-2} + 4S(n-2) = n2^{n-3} \text{ where } S(1) = S(2) = 1.$$

We count 2^{n-2} of the tilings directly and then find a one-to-one correspondence between the remaining tilings and one quarter of the $(n-2) \times (n-2)$ tilings with

$n - 2$ monomers. The even and odd cases have very similar proofs and as such we prove only the even case here.

The trivial case for even n , shown in Fig. 5(a) for $n = 8$, is a horizontal running bond tiling with monomers in the northern corners and $n/2$ monomers on both the West and East boundaries. We break the recurrence into two cases based on what happens to the southern-most (yellow) monomer on each of these boundaries. We will call these monomers w and e respectively.

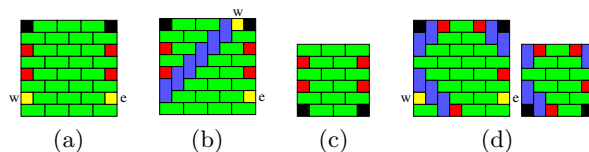


Fig. 5. (a) Trivial case for an 8×8 square with 8 monomers. (b) Flipping a southern-most (yellow) monomer northward. (c) Trivial case for a 6×6 square with 6 monomers. (d) An 8×8 tiling with its associated 6×6 tiling.

The diagonal slices corresponding to w and e can be flipped in a northerly or southerly direction, or not flipped. If w (e) is flipped in a northerly direction, no orthogonal diagonal slices can be flipped (including that corresponding to e (w)). Further, each of the remaining $n - 3$ non-fixed monomers can be independently flipped in the same orientation (though not necessarily the same direction) as w (e), as shown in Fig. 5(b). This gives 2^{n-3} possibilities for each of w and e , resulting in a total of 2^{n-2} tilings.

If w or e are flipped in a southerly direction, they can be flipped independently of each other and of other non-fixed monomers. Thus for each tiling which has w and e fixed, there are four corresponding tilings. As such, we now fix w and e in the trivial tiling as well as the already-fixed northern corner (black) monomers. To count the remaining tilings, we take an inductive approach by finding a one-to-one correspondence between the remaining $n \times n$ tilings and one quarter of the $n - 2 \times n - 2$ tilings with $n - 2$ monomers.

There are $n - 4$ (red) monomers on the East and West edges of the trivial case that we have not fixed. Consider the 180 degree rotation of the trivial case for the $n - 2 \times n - 2$ tilings with $n - 2$ monomers, which has fixed (black) monomers in the two southern corners, as shown in Fig. 5(c) for $n - 2 = 6$. Associate the $n - 4$ non-fixed monomers of this tiling with the $n - 4$ non-fixed monomers of the $n \times n$ trivial case in a natural way: with those in the same relative position paired.

There are $S(n - 2)$ ways of flipping the monomers of the $n - 2 \times n - 2$ trivial case, and thus $S(n - 2)$ ways of flipping the corresponding monomers of the $n \times n$ trivial case. Intuitively it is easy to see that this works. To be rigorous, however, we need to show that if two diagonal slices cannot both be flipped in the larger square, the same is true for the corresponding slices in the smaller square, and vice versa. To do so, observe that the conflict between two flips depends entirely on the distance of the monomers from the horizontal centerline of the trivial tiling.

For any pair of monomers on opposite edges, let d_W and d_E be the distances from the horizontal centerline, with negative distance below the line and positive distance above. If $d_W + d_E > 0$ ($d_W + d_E < 0$) then the two monomers cannot both be flipped southward (northward). This distance from the centerline is preserved between the associated monomers in the larger and smaller squares and thus the compatibilities between flips of monomers are also preserved.

This completes the recurrence defined at the beginning of the proof and establishes the theorem for even n . $\square$

3.2 Fixed Height Tatami Tilings

In this section we show that for a fixed number of rows r , the ordinary generating function of the number of tilings of an $r \times c$ rectangle is a rational function. We show that, for each value of r , the number of fixed-height tilings satisfies a system of linear recurrences with constant coefficients. We will derive the recurrences for small values of r and then discuss an algorithm which can be used for larger values of r .

Let $T_r(z)$ denote the ordinary generating function $T_r(z) = \sum_{c \geq 0} T(r, c)z^c$.

Theorem 3. *For $h = 1, 2, 3$ then generating functions $T_h(z)$ are*

$$T_1(z) = \frac{1+z}{1-z-z^2}, \quad T_2(z) = \frac{1+2z^2-z^3}{1-2z-2z^3+z^4}, \quad \text{and}$$

$$T_3(z) = \frac{1+2z+8z^2+3z^3-6z^4-3z^5-4z^6+2z^7+z^8}{1-z-2z^2-2z^4+z^5+z^6}.$$

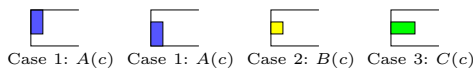
Proof. We only provide a proof outline for height 3. In this proof we focus on explaining the system of recurrences and leave out the technical details of determining the generating function which can be carried out using Maple. We begin by considering the possible unique left-hand side starts of a tatami tiling in a $3 \times c$ rectangle.

Case 1: The tiling begins with a dimer placed vertically in the first column. There are actually two ways to do this since the vertical tile could be placed in the first two rows or the second two rows. We will just count it as one for now and double the result from this case in the end. Let $A(c)$ be the number of $3 \times c$ tilings associated with each of these cases.

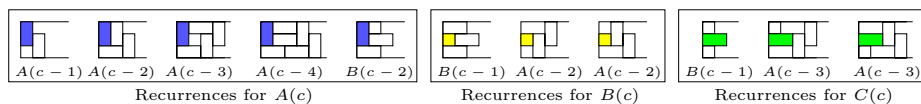
Case 2: The tiling begins with a monomer tile placed in the center (row 2). Let $B(c)$ be the number of tilings associated with this case.

Case 3: The tiling begins with a dimer tile placed horizontally in the center (row 2). Let $C(c)$ be the number of tilings associated with this case.

The following diagram illustrates the three cases.



These cases are mutually exclusive and exhaustive, thus $T(3, c) = 2A(c) + B(c) + C(c) - 3$. We subtract 3 since for each case we define the number of empty tilings to be one so this is tiling is counted 4 times. Now we determine recurrences for each of $A(c)$, $B(c)$ and $C(c)$.



By consideration of the various cases that can occur, as illustrated in the diagram above, we are led to the following three recurrence relations:

$$\begin{aligned} A(c) &= A(c-1) + A(c-2) + A(c-3) + A(c-4) + B(c-2) \\ B(c) &= B(c-2) + 2A(c-2) \quad \text{and} \quad C(c) = B(c-1) + 2A(c-3). \end{aligned}$$

Let $\mathcal{A}(z)$, $\mathcal{B}(z)$, and $\mathcal{C}(z)$ denote the generating functions for $A(c)$, $B(c)$ and $C(c)$ respectively. Using the initial conditions $A(0) = 1, A(1) = 1, A(2) = 3, A(3) = 7, B(0) = 1, B(1) = 1, B(2) = 1, C(0) = 1, C(1) = 0, C(2) = 4$ we can solve for $\mathcal{A}(z)$, $\mathcal{B}(z)$, and $\mathcal{C}(z)$, eventually obtaining the expression for $T_3(z)$ given in the statement of the Theorem. $\square$

For larger values of r the number of cases increases rapidly so we develop an algorithm to aid in determining the recurrences. For the $r = 3$ case discussed above we wanted to have the least number of cases possible to help simplify the calculations.

We begin by considering all options of tiling the first column with the added allowance that horizontal tiles may lie partially in the second column. We store the placement of the tiles for each of the tilings of the first column. Each of these tilings will correspond to a different case in our system of linear recurrences. In the next step we attempt to tile the second column for each way that we tiled the first, again allowing horizontal tiles into the next column. The word “attempt” is used because not all of the first column tile configurations permit a valid tatami tiling of the second column. For each second column tiling the algorithm then runs a check against the stored cases to define the recurrence.

In essence, the first column tilings represent a set of boundary configurations, say S , and for each element in S we set up a recurrence by matching each of its second column boundary configurations with an element in S . For initial conditions we determine the number of ways to complete an r by 2 rectangle for each first column tiling. The system of recurrences is then solved using Maple. Note that the algorithm produces a system of linear recurrences with constant coefficients which implies the generating functions for the system will be rational functions. Hence, for a fixed number of rows r , the generating function for the number of tatami tilings of an $r \times c$ rectangle is a rational function. The output of our algorithm for $r = 10$ has degree 65 in the numerator and degree 56 in the denominator.

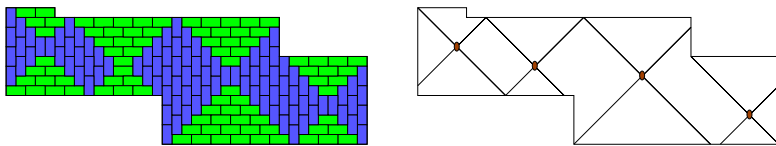


Fig. 6. On the left: The solution to the question posed in Figure 1; no monomers are required to tatami tile the region. On the right: A legal configuration of six magnetic water striders in an orthogonal “pond”. Note that no further striders may be added.

4 Conjectures and Further Research

In this section we list some open problems and conjectures.

- It would be interesting to extend the structural analysis to orthogonal regions. We believe that the main structural components are the same as they were for rectangles, but there are a few subtleties to be clarified at inside corners.
- What is the computational complexity of determining the least number of monomers that can be used to tile an orthogonal region given the segments that form the boundary of the region and the unit size of each dimer/monomer?
- Interpreted as a matching problem on a grid graph G , a tatami tiling is a matching M with the property that $G - M$ contains no 4-cycles. Note that there is always such a matching (e.g., take the “running bond” layout on the infinite grid graph and then restrict it to G). However, if we insist on a perfect matching, then the problem is equivalent to our “perfect” driveway paving problem from the introduction. Thus there are a variety of natural problems about tatami tilings of an arbitrary graph.
- It turns out that the problem of minimizing the number of monomers in a tiling is related to what we call the “magnetic water strider problem”. This time the orthogonal region is a pond populated by water striders. A water strider is an insect that rides atop water in ponds by using surface tension. Its 4 longest legs jut out at 45 degrees from its body. In the fancifully named magnetic water strider problem, we require the body to be aligned north-south. Furthermore its legs support it, not by resting on the water, but by extending to the boundary of the pond. Naturally, the legs of the striders are not allowed to intersect. A legal configuration of magnetic water striders in an orthogonal pond is shown on the right in Fig. 6. Here the problem is a maximization problem, namely what is the largest number of magnetic waters striders that a pond can support?
- Various games would also be of interest. Given an orthogonal region players take turns placing dimers (or dimers and monomers); each placement must satisfy the tatami constraint and the last player who can move wins. A similar game could be played with magnetic water striders. The URL, <http://www.theory.cs.uvic.ca/~cos/oku/> is reserved for such games.

- Given $r + c$ triples of numbers (h, v, m) , one for each row and one for each column, is there a tatami tiling which has h horizontal dimers, v vertical dimers, and m monomers in the respective row or column? Similar questions are important in tomography [5].

Acknowledgements

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